SNOWFLAKE DATA ENGINEERING PROJECT

Healthcare Intelligence Hub

Patient, Cost & Care Analytics

Phase 1 — Ideation Phase

Institutional template-based phase document

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Prepared from the verified project source (query.sql, Health/streamlit_app.py, Health/snowflake.yml, six CSV datasets). Bracketed fields are placeholders for information not supplied.

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Phase 1 — Ideation Phase

Project: Healthcare Intelligence Hub: Patient, Cost & Care Analytics **Templates covered:** Define the Problem Statements (2 marks) · Empathy Map Canvas (4 marks) · Brainstorm & Idea Prioritization (4 marks)

> **Evidence basis.** Everything in this document was checked against the uploaded project archive (`query.sql` , `Health/streamlit_app.py` , `Health/snowflake.yml` , `Health/pyproject.toml` and six CSV files). Items marked *Proposed* are not present in the code. Team details are placeholders because they were not supplied.

1.1 Ideation Summary

1.1.1 Abstract

Hospitals record every patient visit across several operational sources: patient registration, doctor rosters, facility master data, diagnosis coding and billing. Administrators and finance teams rarely see these together. The Healthcare Intelligence Hub loads six CSV extracts into a Snowflake database (`HEALTH_DB`) and moves them through four schemas: `RAW` , `STAGING` , `CURATED` and `MART` . Along the way it cleans and standardises the data, models it as one visit-level fact table (`FACT_PATIENT_VISITS`) with five dimensions, and exposes ten analytical views. A Streamlit-in-Snowflake application reads the fact table and gives users six KPI cards, eight charts, two tables and five filters. The dataset covers 10,000 visits by 2,889 distinct patients across 10 facilities, 60 doctors and 15 diagnoses, from January 2024 to August 2026.

Keywords: Snowflake, data engineering, medallion layers, star schema, healthcare analytics, Streamlit in Snowflake, readmission rate, length of stay, cost analytics.

1.1.2 Problem statement

> **Hospital administrators and finance teams need a consolidated, queryable view of patient visits — spanning cost, length of stay, satisfaction and readmissions — that today sits fragmented across raw operational records. Without a structured warehouse, it is difficult to answer basic questions such as which facilities and doctors are performing well, which diagnoses drive the longest and most expensive stays, and where patient satisfaction is at risk.**

(Official problem statement from the project workspace.)

The data needed to answer these questions already exists in operational records: patients, doctors, facilities, diagnoses, a calendar, and visit-level billing and outcomes. It is not organised for analysis. Each question today means joining raw extracts by hand, so answers are slow, inconsistent and hard to compare across facilities or doctors. The project's purpose is to build that structured warehouse and put an interactive view on top of it.

1.1.3 Existing challenges in healthcare analytics

Table 1.1 — Challenges identified during ideation

#	Challenge	Effect on decision-makers
1	Data is held in separate sources: patients, doctors, facilities, diagnoses and visits/billing	No single view of a visit with its cost and outcome

#	Challenge	Effect on decision-makers
2	Inconsistent text values (case, stray spaces) between sources	Group-by results split one category into several
3	Manual spreadsheet consolidation	Reports go stale; numbers differ between teams
4	No common KPI definitions (e.g., readmission rate)	Facilities and doctors cannot be compared fairly
5	Cost components (consultation, medication, lab, room) and payer split (insurance and patient) are not analysed together	Finance cannot see what drives cost or out-of-pocket burden
6	Static reports with no filtering	Users cannot narrow by date, facility, specialty, visit type or diagnosis category

1.1.4 Proposed solution and innovation

The proposed solution is a cloud data-warehouse pipeline with an interactive application on top:

1. **Ingest:** a named CSV file format and an internal stage in Snowflake, with typed `*_RAW` tables.
2. **Clean:** `STAGING` tables apply `TRIM`, `INITCAP` and NOT NULL key filters.
3. **Model:** a `CURATED` star schema of five `DIM_*` tables and a visit-grain `FACT_PATIENT_VISITS`.
4. **Serve:** ten `MART` views that hold the KPI logic (executive KPIs, monthly trend, diagnosis, facility, doctor, visit type, treatment, admission and readmission analytics).
5. **Explore:** a Streamlit app that runs inside Snowflake, so data never leaves the platform. It uses the active Snowpark session with no stored credentials and offers five filters.

What makes it innovative: the whole stack (storage, transformation, KPI logic and user interface) runs inside a single governed platform. There is no separate BI server and no credential handling in the app (`get_active_session()`). The layered schemas also make every number traceable back to its raw row.

1.1.5 Target users and stakeholders

Table 1.2 — Stakeholders

Stakeholder	Interest	Primary outputs used
Hospital administrator	Visit volumes, length of stay, facility and doctor workload, satisfaction	KPI cards, monthly trend, facility performance, doctor leaderboard
Finance team	Total cost, cost breakdown, insurance coverage and patient payment	KPI view totals, cost-breakdown chart, facility cost table
Clinical quality lead	30-day readmissions, treatment outcomes	Readmission-by-diagnosis chart, treatment status chart
Data engineer	Reliable, auditable pipeline	RAW, STAGING, CURATED and MART schemas; validation queries
Faculty evaluator	Correctness and completeness of the project	This documentation set

1.1.6 Business value and expected benefits

- One trusted, queryable source for visit, cost and outcome metrics.
- KPI definitions written once in SQL views and reused, instead of being recalculated in spreadsheets.
- Faster ad-hoc analysis: filters and charts recompute in the app, with no export step.

- A basis for quality improvement. For example, the analysis shows Depression (14.05%) and Back Pain (13.98%) with the highest readmission rates, against 12.03% overall.

1.1.7 Scope, assumptions, constraints and feasibility

In scope: the six supplied CSV files, the Snowflake objects in `query.sql`, the Streamlit app in `Health/`, data-quality checks, KPIs and dashboards.

Out of scope: live EHR/HIS integration, real-time streaming, predictive models, patient-facing features, and production RBAC (proposed only).

Assumptions: - The CSV data is synthetic or anonymised. Names repeat across records (20 first names and 15 surnames across 3,000 patients), which suggests generated data. - Monetary values share one currency. The source does not name it, and the app labels values with `$`. - `READMISSION_WITHIN_30_DAYS` is pre-computed upstream.

Constraints: a Snowflake account with a warehouse (`COMPUTE_WH`); Streamlit in Snowflake with the packages its runtime allows; academic timeline.

Feasibility:

Dimension	Assessment
Technical	**High.** All required objects are standard Snowflake SQL. The app uses `streamlit`, `pandas`, `altair` and Snowpark, all available in Streamlit in Snowflake.
Data	**High.** The data is complete: no NULL or duplicate keys, and every foreign key resolves (verified, see Phase 6).
Economic	**High.** A data volume of about 1.4 MB of CSV fits comfortably in an X-Small warehouse or a trial account.
Operational	**Medium.** Loading is a manual step today, and scheduling and RBAC are proposed.

1.1.8 Proposed technology stack (verified)

Layer	Technology	Verified in
Storage / warehouse	Snowflake (`HEALTH_DB`; schemas `RAW`, `STAGING`, `CURATED`, `MART`)	`query.sql`
Ingestion	Snowflake file format `RAW.CSV_FORMAT` and internal stage `RAW.HEALTHCARE_STAGE`	`query.sql`
Transformation	Snowflake SQL (CTAS, views)	`query.sql`
Application	Streamlit in Snowflake, Python 3.11, pandas, Altair, Snowpark	`Health/*.py`, `pyproject.toml`, `snowflake.yml`
Deployment descriptor	Snowflake CLI project definition v2 (`snowflake.yml`)	`Health/snowflake.yml`

1.1.9 Expected outcomes and success criteria

#	Success criterion	Measure	Status
SC-1	All six sources load into RAW with no row loss	RAW count = CSV rows	Verified locally (Phase 6, TC-ING-01…06)
SC-2	Zero NULL or duplicate business keys after staging	Null/duplicate checks = 0	Verified locally

#	Success criterion	Measure	Status
SC-3	Every visit links to patient, doctor, facility, diagnosis and date	Orphan count = 0	Verified locally
SC-4	KPIs reproducible from source	View KPIs = independent pandas result	Verified locally (TC-KPI-01…06)
SC-5	Dashboard responds to all five filters	Filtered counts correct	Logic verified locally; live app **Not Executed**

1.2 Define the Problem Statements

Field	Value
Date	31 January 2026 *(as printed on template)*
Team ID	[Team ID]
Project Name	Healthcare Intelligence Hub: Patient, Cost & Care Analytics
Maximum Marks	2 Marks

Customer Problem Statement Template: Create a problem statement to understand your customer's point of view.

Table 1.3 — Customer problem statements

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a hospital administrator responsible for 10 facilities and 60 doctors	see which facilities and doctors are performing well on visit volume, cost, length of stay and satisfaction	visit, doctor and facility data sit fragmented across raw operational records that I must join by hand	there is no structured warehouse linking each visit to its doctor and facility	frustrated and unsure whether the comparisons I present are correct
PS-2	a finance team member	find which diagnoses drive the longest and most expensive stays, and how that cost is split between insurance and the patient	length of stay and cost components are buried in visit-level records with no diagnosis view	no consolidated, queryable model aggregates cost and length of stay by diagnosis	under pressure, because each cost review takes days to rebuild
PS-3	a hospital administrator responsible for patient experience	find where patient satisfaction is at risk, and where readmissions are high, by facility, doctor and diagnosis	satisfaction scores and readmission flags are recorded per visit but never summarised	there is no shared KPI definition or filterable view	worried that dissatisfied patients and preventable readmissions go unnoticed

1.3 Empathy Map Canvas

Field	Value
Date	31 January 2026 *(as printed on template)*

Field	Value
Team ID	[Team ID]
Project Name	Healthcare Intelligence Hub: Patient, Cost & Care Analytics
Maximum Marks	4 Marks

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes. The persona here is a **hospital administrator / finance manager** (combined persona). The entries are design assumptions derived from the project objectives, not interview transcripts.

Table 1.4 — Empathy map: Hospital administrator / finance manager

SAYS	THINKS
"I need one number for total visits that everyone agrees on." "Which facility is the most expensive per visit?" "Why are readmissions going up?"	Are my spreadsheets joined correctly? Is a cost increase driven by room charges or by medication? Are some doctors overloaded?
DOES	**FEELS**
Exports CSVs from separate systems every month Uses VLOOKUP to combine visits with doctor and facility names Builds static charts for review meetings	Overwhelmed by manual work Anxious about presenting wrong figures Wants confidence and control
PAINS	**GAINS**
Inconsistent category labels Slow turnaround No drill-down by date, facility, specialty or diagnosis	One consolidated, governed dataset KPIs refresh with filters in seconds Comparable facility, doctor and diagnosis metrics

1.4 Brainstorm & Idea Prioritization

Field	Value
Date	31 January 2026 *(as printed on template)*
Team ID	[Team ID]
Project Name	Healthcare Intelligence Hub: Patient, Cost & Care Analytics
Maximum Marks	4 Marks

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving.

Step 1: Brainstorm and Idea Listing

Table 1.5 — Brainstormed ideas

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	[Team Member 1]	Load all six CSVs into Snowflake through a stage and file format	Data ingestion	1
2	[Team Member 2]	Separate RAW, STAGING, CURATED and MART schemas for traceability	Data architecture	1

S.No	Team Member	Idea / Suggestion	Category	Group No.
3	[Team Member 3]	Standardise text with TRIM/INITCAP and drop rows with NULL keys	Data quality	1
4	[Team Member 4]	Star schema: one visit fact plus patient, doctor, facility, diagnosis and date dimensions	Data modelling	2
5	[Team Member 1]	SQL views for KPIs: visits, cost, LOS, satisfaction, readmission rate	Analytics	2
6	[Team Member 2]	Interactive Streamlit app inside Snowflake with sidebar filters	Visualisation	3
7	[Team Member 3]	Tableau Public dashboard on extracted data	Visualisation	3
8	[Team Member 4]	Predict readmission risk with an ML model	Advanced analytics	4
9	[Team Member 1]	Automate refresh with Snowflake Streams and Tasks	Automation	4
10	[Team Member 2]	Role-based access and masking of patient names	Security	4

Step 2: Idea Prioritization

Table 1.6 — Idea prioritisation

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority	Selected (Yes/No)
1	Layered Snowflake ingestion and cleaning pipeline (RAW → STAGING)	High	High	1	Yes (implemented)
2	Star-schema CURATED layer and MART KPI views	High	High	2	Yes (implemented)
3	Streamlit-in-Snowflake interactive dashboard (chosen over Tableau, to keep data and app in one platform)	High	High	3	Yes (implemented)
4	Automation, RBAC/masking and ML readmission prediction	Medium	Medium	4	No: deferred to future scope (Phase 8)